

the value of the minimum trade risk quantity, which makes sense since the residual risk curve is symmetric around the minimum value.

If traders need to adhere to the shrinking portfolio constraint the bounds on z_k are:

$$0 \leq z_k \leq r_k$$

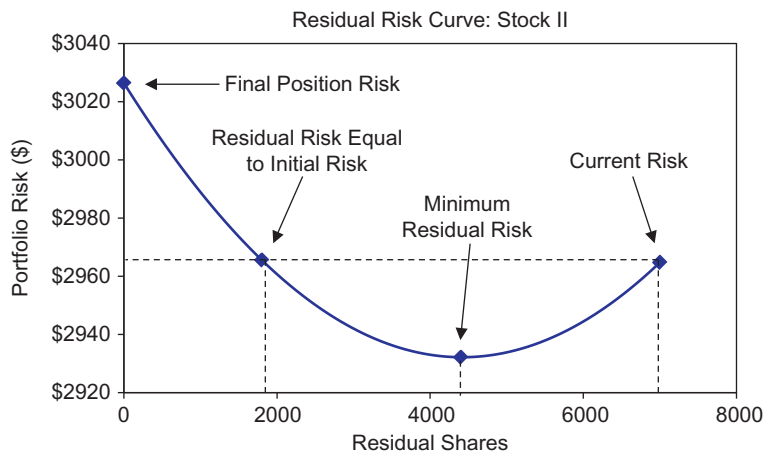
Thus the actual number of shares that can be traded adhering to the shrinking portfolio constraint is:

$$z_k^* = \min(\max(0, z_k), r_k)$$

Figure 9.5 depicts the residual risk curve for stock II. At the initial portfolio position there are $r_2 = 7000$ shares of stock II and the total portfolio risk is \$2965. Total risk can be minimized by trading $\gamma_2 = 2583$ shares resulting in total risk of \$2,932 and a new position size for stock II of 4417 shares. Traders can transact up to $z_2 = 5167$ shares, resulting in a new position size of stock II of 1833 shares, and still have the same overall risk of \$2965 as we had prior to trading. Trading more than 5167 shares would result in higher risk exposure.

When to Use These Values?

The minimum trading risk and maximum trading opportunity quantities provide valuable guidelines for how much an algorithm or trader can



■ Figure 9.5 Residual Risk Curve Stock II.